

# HANRONG ZHANG

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## EDUCATION

### University of Illinois Chicago

Ph.D. Student in Computer Science

Big Data and Social Computing Lab; Supervisor: Prof. Philip S. Yu

Aug. 2025 – Present

Chicago, USA

### Zhejiang University

MEng. Computer Engineering

Ranking: 1/82; GPA: 93/100; Supervisor: Prof. Hongwei Wang

Sep. 2022 – Mar. 2025

Hangzhou, China

### University of Leeds

BSc. Computer Science - First Class Honors Degree

Sep. 2018 – Jun. 2022

Leeds, United Kingdom

### Southwest Jiaotong University

BEng. Computer Science and Technology

Ranking: 1/74; GPA: 3.81/4.0, 92/100; Supervisor: Prof. Tianrui Li

Sep. 2018 – Jun. 2022

Chengdu, China

## INTERNSHIP

### Google LLC

Student Researcher, Data Synthesis for Gemini and Long-Horizon GUI Agents

Mountain View, California

May 2026 – Present

### Alibaba Group

Research Intern, RL for Human-Agent Multi-turn Interaction

Hangzhou, Zhejiang, China

May 2025 – Aug. 2025

- Build scalable simulation environments for LLM-human multi-turn interactions, covering tool-call scenarios such as retail and airline services under sparse rewards, limited data, and unstable environments.
- Propose a **Tool Dependency Graph** to model tool constraints and generate high-quality multi-turn tool-call dialogues with ground-truth trajectories for verifiable reward computation. Train agents with synthesized trajectories via multi-turn **GRPO**, improving long-horizon tool use in human-agent interactions.

## SELECTED PUBLICATIONS AND PREPRINTS

Full publication list is available on [Google Scholar](#). \* denotes Equal Contribution. † denotes Corresponding Author.

- [1] **Hanrong Zhang**, Jingyuan Huang, Kai Mei, Yifei Yao, Zhenting Wang, Chenlu Zhan, Hongwei Wang, Yongfeng Zhang, *Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents*, **ICLR 2025**. [Paper] [Code] [Website] (A foundational benchmark for LLM Agent Security, 420+ citations & 290+ GitHub stars)
- [2] **Hanrong Zhang**, Zhenting Wang, Tingxu Han, Mingyu Jin, Chenlu Zhan, Mengnan Du, Hongwei Wang<sup>†</sup>, Shiqing Ma, *Invisible Backdoor Attack in Self-supervised Learning*, **CVPR 2025**. [Paper] [Code] (Machine Learning Safety)
- [3] **Hanrong Zhang**, Shicheng Fan, Henry Peng Zou, Yankai Chen<sup>†</sup>, Zhenting Wang, Jiayu Zhou, Chengze Li, Wei-Chieh Huang, Yifei Yao, Kening Zheng, Xue Liu, Xiaoxiao Li, Philip S. Yu, *CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification*, **COLM 2026**. [Paper] [Code] [Website] **Invited Talk**: Alibaba Group, *Towards Self-Evolving Agent Skills*. (Agent Skill Self-Evolution, 82+ citations)
- [4] Zhaofen Wu\*, **Hanrong Zhang**\*<sup>†</sup>, Fulin Lin, Wujiang Xu, Xinran Xu, Yankai Chen, Henry Peng Zou, Shaowen Chen, Weizhi Zhang, Xue Liu, Philip S. Yu, Hongwei Wang<sup>†</sup>, *GAM: Hierarchical Graph-based Agentic Memory for LLM Agents*, **ACL 2026**; ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (**Oral**). [Paper] (Agent Memory)
- [5] **Hanrong Zhang**, Yifei Yao, Zixuan Wang, Jiayuan Su, Mengxuan Li, Peng Peng, Hongwei Wang, *Class Incremental Fault Diagnosis under Limited Fault Data via Supervised Contrastive Knowledge Distillation*, **IEEE Transactions on Industrial Informatics**. (IF=12.3, SCI Q1, Top Journal) [Paper] (Continuous Learning)
- [6] Yifei Yao\*, **Hanrong Zhang**\*, Fulin Lin\*, Ziyang Jin, Xiaoxiao Li, Hongwei Wang, Ying Chi, *From Uncertainty to Clarity: Uncertainty-Guided Class-Incremental Learning for Limited Biomedical Samples via Semantic Expansion*, **IEEE Journal of Biomedical and Health Informatics**. (IF=7.7, SCI Q1) [Paper] (Continuous Learning)
- [7] Xingyue Wang\*, **Hanrong Zhang**\*, Xinlong Qiao, Ke Ma, Shuting Tao, Peng Peng, Hongwei Wang, *Generalized Out-of-distribution Fault Diagnosis (GOOFD) via Internal Contrastive Learning*, **IEEE Transactions on Industrial Informatics**. (IF=12.3, SCI Q1, Top Journal) [Paper] (OOD Distribution Detection)

- [8] **Hanrong Zhang**, Yankai Chen<sup>†</sup>, Shicheng Fan, . . . , Dawn Song, Philip S. Yu, Xue Liu, *Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey*, **Preprint**. [Paper] (Agent Learning & Data Synthesis)
- [9] **Hanrong Zhang**, Xinyue Wang, Jiabao Pan, Hongwei Wang, *SAKA: an intelligent platform for semi-automated knowledge graph construction and application*, **Service Oriented Computing and Applications**. [Paper] (Knowledge Graph)
- [10] Zhiling Yan, Dingjie Song, **Hanrong Zhang**, Wei Liang, Yuxuan Zhang, Yutong Dai, Lifang He, Philip S. Yu, Ran Xu, Xiang Li, Lichao Sun, *OpenSkill: Open-World Self-Evolution for LLM Agents*, **EMNLP 2026**. [Paper] [Code] (Agent Self-Evolution)

## SELECTED HONORS

|                                                                 |                                            |
|-----------------------------------------------------------------|--------------------------------------------|
| Outstanding Graduate in Zhejiang Province & Zhejiang University | Zhejiang University, 2025                  |
| National Scholarship for Graduate Students (Top 0.2%)           | Zhejiang University, 2023 - 2024           |
| National Scholarship for Graduate Students (Top 0.2%)           | Zhejiang University, 2022 - 2023           |
| National Scholarship for Undergraduate Students (Top 0.2%)      | Southwest Jiaotong University, 2019 - 2020 |
| Outstanding Graduate in Sichuan Province                        | Southwest Jiaotong University, 2022        |
| Best Student in Computer Science (1/75)                         | University of Leeds, 2020 - 2021           |
| First-class full-ride Scholarship (1/75)                        | University of Leeds, 2020 - 2021           |
| Best Student Overall (1/300, 4 majors)                          | University of Leeds, 2018 - 2019           |

## RESEARCH EXPERIENCE

### Agent Security Bench (ASB)

ICLR 2025, First Author

- **Motivation:** LLM agents extend LLM reasoning into real-world interaction: by calling tools, updating memory, and executing multi-step plans, they can change external environments, making agent security more critical than standard chatbot safety; however, prior evaluations lacked a unified benchmark across prompts, tools, memory, and planning.
- **Contribution:** Introduce the first LLM agent security benchmark with 10 scenarios, 10 agents, 400+ tools, 27 attack/defense methods, 7 metrics, and Net Resilient Performance (NRP) for balancing utility and security.
- **Results:** Reveal that current LLM agents remain highly vulnerable and existing defenses provide limited protection: attacks reach up to 84.30% success across prompt, tool, and memory stages, while defenses struggle to preserve both utility and security across 13 LLM backbones and 90,000+ test cases.

### Towards Self-Evolving Skills in LLM Agents

CoEvoSkills (COLM 2026) & OpenSkill (EMNLP 2026)

- **Motivation:** Skills can boost professional tasks, but human-curated skills are costly and can even hurt performance in some SkillsBench domains. Meanwhile, one-shot self-generation methods are nearly useless, and prior evolution methods are mostly designed for tools and cannot reliably build structured multi-file skill packages.
- **Contribution:** Propose **CoEvoSkills** for closed-world skill evolution and **OpenSkill** for open-world self-evolution, refining multi-file skills through surrogate verification, open-world knowledge retrieval.
- **Results:** **CoEvoSkills** reaches average 71.1% Pass rate on SkillsBench (+40.5pp over no-skill, +17.6pp over human-curated skills baselines); **OpenSkill** reaches 43.6% on Claude Opus 4.6 and 42.1% on GPT 5.2, within 1-3pp of human-curated skills and about +9pp over the best closed-world baselines.

### GAM: Hierarchical Graph-based Agentic Memory for LLM Agents

ACL 2026, Co-First & Corresponding Author

- **Motivation:** Long-term agent memory faces a core conflict between fast contextual adaptation and stable knowledge retention: stream-based memories update quickly but lack write isolation, causing memory loss and semantic drift, while static graph memories preserve structure but struggle with fluid open-domain dialogue.
- **Contribution:** Propose **GAM**, a hierarchical graph memory framework that decouples memory encoding from consolidation: an Event Progression Graph isolates ongoing dialogue events, while semantic-boundary triggers consolidate complete episodes into a Topic Associative Network for stable long-term knowledge.
- **Results:** **GAM** achieves SOTA performance among six memory baselines on LoCoMo and LongDialQA.

## OPEN-SOURCE PROJECTS

### Dr. Claw [Paper] [Code]

EMNLP 2026 · Second Author · Core Contributor

- Build an **Auto Research** workspace that orchestrates agentic pipelines from research ideation and literature/code survey to experiment execution, analysis, paper writing, and presentation generation.
- Integrate multi-source research news tracking, 100+ research skills, project/session management, and multi-agent backends into a full-stack research IDE with 1,000+ GitHub stars and 100+ forks.

## ACADEMIC SERVICE

Conference Reviewer: AAAI 2027, ICLR 2026-2027, ICML 2026, NeurIPS 2026, CVPR 2026, ECCV 2026, KDD 2026, MICCAI 2026

Journal Reviewer: TPAMI, TNNLS, TMLR, Pattern Recognition